# AtlasSAT-16U Souss Water Observation CubeSat

Independent Research Project | Document 2A | Orbital Analysis | Rev D3.4
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16U CubeSat · 550 km near-SSO · GMAT R2026a
VNIR pushbroom · 30 km nominal swath
Chtouka-Massa agriculture, south of Agadir

## ENGINEERING OBJECTIVE

Test whether one AtlasSAT-16U can offer agricultural targets south of Agadir a **daylight opportunity every** $\leq 72$ **h**. Preserved GMAT dynamics, provisional GIS targets and checked optical-access calculations form the evidence.

## PRESERVED CONCEPT

| Model | Adopted input |
|---|---|
| Orbit | Near-SSO, 550 km concept |
| Semimajor axis | 6928.1363 km |
| Ecc. / incl. | $e$=0.001; $i$=97.5° |
| RAAN | 6.463° |
| AOP / TA | 0° / 0° (canonical) |
| Epoch (UTC) | 20 Oct 2026, 12:00:00.000 |
| Initial axes | EarthMeanOfDate (Earth; MODEq) |
| LTDN / span | 10:30 mean solar; 30 days |
| Payload | Five-band VNIR pushbroom |
| Swath / roll | 30 km; nadir, ideal $\pm 10°$ and $\pm 20°$ |
| Daylight | Target Sun elevation $\geq 0°$ |

## WHAT WAS VERIFIED

Archived orbit benchmark: **36 numerical checks**. Coverage geometry: **570 opportunities at 31 targets**; maximum boundary difference **1.197** $\mu$**s**. Numerical consistency is not physical accuracy.

## FOUR EVIDENCE LEVELS

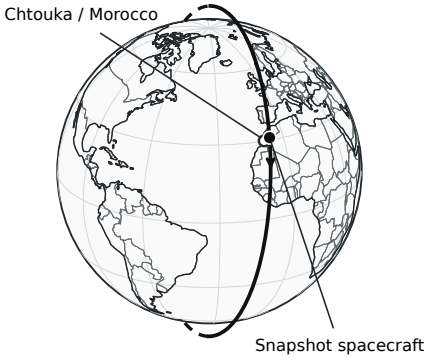
**Passage:** a track crosses the region.

**Reachability:** a target lies inside an ideal sensor envelope.

**Daylight opportunity:** reachability plus illumination.

**Physical validation:** an achievable imaging service remains unproven.

## PRESERVED NEAR-SSO GEOMETRY




1 Nov 2026, 11:23:49 UTC snapshot. True orbit-to-Earth scale; solid near-side / dashed far-side arc. Predicted orientation. Natural Earth context.

## CHTOUKA–MASSA AGRICULTURAL TARGET




Seven-commune study boundary
Primary agricultural mask (provisional)
Major roads (OSM)
Greenhouse-source southern edge

*Morocco locator: geographic context only.
WGS84 / UTM 29N; grid-north arrow.

**396.47 km**$^2$ primary mask in a **1,173.92 km**$^2$ seven-commune envelope. **Provisional research ROI**, not an official agricultural perimeter.

## KEY RESULTS | DAYLIGHT | 30 km

| Metric | Nadir | $\pm 10°$ | $\pm 20°$ |
|---|---|---|---|
| Mean 72 h area | 11.48% | 22.76% | 33.44% |
| Max complete gap | 360.01 h | 360.01 h | **312.22 h** |
| Largest span* | 720 h | 360.01 h | 312.22 h |

*Includes endpoint censoring. Hourly window starts; 250 m area quadrature. Opportunities, not acquired images.

## ENGINEERING SIGNIFICANCE

Slow ground-track drift clusters opportunities rather than providing daily service. A favourable mean cannot replace a maximum-gap test; a night pass cannot meet daylight needs. The analytical note checks this interpretation; future design trades remain unperformed.

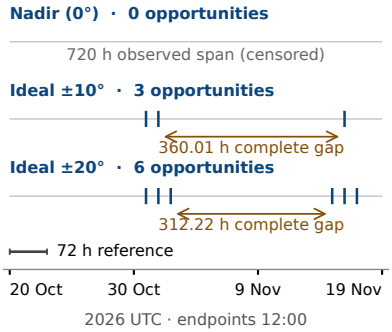
## DAYLIGHT REVISIT VERDICT

**NOT MET**

**312.22 h** vs **72 h**

Longest complete gap with ideal $\pm 20°$ roll. All 14,494 samples fail the 72 h test.

## 30-DAY DAYLIGHT ACCESS TIMELINE




30 km; Sun $\geq 0°$. Ticks: point opportunities, not acquired images or time-scaled bars. E16.

## TRACEABLE WORST CASE

Sidi Bibi target `sidi-bibi_-10750_19500`; 2–15 November. REF lies outside the ROI and is **excluded**. Full coordinates and UTC endpoints: 2A-1 / E16.

## BOUNDARIES OF THE EVIDENCE

- **Provisional** agriculture; local classification accuracy unestablished.
- Greenhouse-source gap: **22.04%** of envelope / **6.861%** of mask. Missing acreage unknown.
- **Predicted** Earth orientation; preliminary forces.
- Ideal roll is not qualified slew, pointing or image quality; physical/operational acquisition unvalidated.
- TA90 changes initial TA at fixed osculating inputs, not phase alone at fixed mean orbit.

## NEXT STEPS

Next: phased 2–5 satellite constellation trades, followed by orbit-design, drag, solar-elevation and observation-scheduling studies.

## DETAILED ENGINEERING RECORD

2A-1, Rev D3.5: methods and gap provenance. The GMAT model, Python examples and evidence index are available in the Study 2A repository.

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